With this definition, the flux is consistent and conservative. We may write

$$\begin{aligned}\hat{f}(u_i, u_k; n_{ik}) &= f(u_i)\cdot n_{ik} + \frac{1}{2}(f(u_k) - f(u_i))\cdot n_{ik} - \frac{\alpha_{ik}}{2}(u_k - u_i) \\ &= f(u_i)\cdot n_{ik} + \frac{1}{2}\left(\overline{a}_{ik} - \alpha_{ik}\right)(u_k - u_i) \end{aligned} \tag{9.31}$$

Note that $(f(u_k) - f(u_i))\cdot n_{ik} = \overline{a}_{ik}(u_k - u_i)$ holds by definition (9.30). (If $u_i = u_k$ this holds trivially as long as the coefficients $\overline{a}_{ik}$ remain finite. This is also ensured by (9.30).) Alternatively, isolating $f(u_k)$, we may write the flux as

$$\begin{aligned}\hat{f}(u_i, u_k; n_{ik}) &= f(u_k)\cdot n_{ik} - \frac{1}{2}(f(u_k) - f(u_i))\cdot n_{ik} - \frac{\alpha_{ik}}{2}(u_k - u_i) \\ &= f(u_k)\cdot n_{ik} - \frac{1}{2}\left(\overline{a}_{ik} + \alpha_{ik}\right)(u_k - u_i) \end{aligned} \tag{9.32}$$

If (9.29) holds, it is obvious from eq. (9.31) that $\hat{f}$ is monotonically decreasing in its second argument, and from (9.32) it is clear that it is monotoncially increasing in its first argument. Hence, the flux is monotone in the sense of Definition 9.3. We can write the scheme in the from (9.20), with positive coefficients. To make this explicit, substitute (9.31) into (9.28) to obtain

$$\frac{du_i}{dt} = \frac{1}{|T_i|}\sum_{k\in\mathcal{N}_i}\frac{1}{2}\left(\alpha_{ik} - \overline{a}_{ik}\right)(u_k - u_i)|S_{ik}|.$$

(Note that the first term in (9.31) vanishes in the summation over a closed surface.) Clearly the coefficients $C_{ik} = 1/2\left(\alpha_{ik} - \overline{a}_{ik}\right)|S_{ik}|$ are positive under the constraint (9.29). We can conclude that, for u_i a local minimum (respectively maximum) $\frac{du_i}{dt} \geq 0$ (respectively $\frac{du_i}{dt} \leq 0$). The fully discrete scheme can be made stable under the simple time step restriction as in Theorem 9.2.

9.5 The Lax-Wendroff Theorem

For multidimensional nonlinear systems of hyperbolic conservation laws, the well-posedness theory is highly deficient. Naturally then, there is no complete convergence theory for numerical schemes either. For scalar equations (which have received most of our attention so far), there is both a complete well-posedness theory, and a convergence theory for the finite-volume schemes which we have considered in this chapter. However, a full account of this theory is beyond the scope of these introductory notes. We restrict ourselves again to the one-dimensional case, and give a useful theorem, that states that a convergent conservative scheme must necessarily converge to a proper weak solution. This is the content of the *Lax-Wendroff theorem.*

In the following we extend the numerical solution to a function defined for all $(x,t) \in \mathbb{R}\times(0,\infty)$ in a piecewise constant manner:

$$u_h(x,t) := u_i^n, \quad \text{for } x_{i-\frac{1}{2}} < x < x_{i+\frac{1}{2}}, \quad t^n < t < t^{n+1}$$

Theorem 9.3 (Lax-Wendroff) *Assume that the one-dimensional scheme (Scheme 9) with forward difference time stepping (eq.* (9.8)*), is consistent with the conservation law* (9.1)*, and let $\tau = \mathcal{O}(h)$. Assume that there exists a constant C, such that*

$$\sup_{\substack{i\in\mathbb{Z}\\ n\geq 0}}|u_i^n| = \sup_{(x,t)\in\mathbb{R}\times(0,\infty)}|u_h(x,t)| \leq C$$

uniformly in h, and that for some function u,

$$\lim_{h\to 0}\int_K |u_h - u|dx = 0 \tag{9.33}$$

for all compact subsets $K \subset \mathbb{R} \times (0, \infty)$. Then u is a weak solution of (9.1), i.e. u satisfies

$$\int_0^\infty \int_{\mathbb{R}} u v_t + f(u) v_x \, dxdt + \int_{\mathbb{R}} u_0(x) v(x,0) \, dx = 0, \qquad \forall v \in C_0^1(\mathbb{R} \times [0,\infty)) \tag{9.34}$$

Proof (Sketch) Choose an arbitrary test function $v \in C_0^1(\mathbb{R}\times[0,\infty))$, and set $v_i^n := v(x_i, t^n)$. Multiply (9.8) by $h\,\tau\, v_i^n$, and sum over all i, n, to obtain

$$h \sum_{i\in\mathbb{Z}} \sum_{n=0}^{\infty} v_i^n \left(u_i^{n+1} - u_i^n\right) + \tau \sum_{i\in\mathbb{Z}} \sum_{n=0}^{\infty} v_i^n \left(\hat{f}_{i+\frac{1}{2}}^n - \hat{f}_{i-\frac{1}{2}}^n\right) = 0.$$

Note that the infinite sums do not present a problem here, due to the compact support of the test functions: The sums may be truncated to finite sums that cover the support of v. Rearranging the sums, and multipying by "-1", we have

$$h \sum_{i\in\mathbb{Z}} \sum_{n=0}^{\infty} u_i^{n+1} \left(v_i^{n+1} - v_i^n\right) + \tau \sum_{i\in\mathbb{Z}} \sum_{n=0}^{\infty} \hat{f}_{i+\frac{1}{2}}^n \left(v_{i+1}^n - v_i^n\right) + h \sum_{i\in\mathbb{Z}} u_i^0 v_i^0 = 0.$$

Define v_h as a piecewise constant function in the same manner as u_h, and define $\hat{f}_h$, such that

$$\hat{f}_h = \hat{f}_{i+\frac{1}{2}}, \qquad \text{for } x_i < x < x_{i+1}, \quad t^n < t < t^{n+1}.$$

(Note that the piecewise constant flux function is defined with an offset of $h/2$ compared to the u_h and v_h.) Now, noting that for any of the piecewise constant functions defined thus far, e.g. for u_h, we have

$$h\tau \sum_{i,n} u_i^n = \int_0^\infty \int_{\mathbb{R}} u_h \, dxdt,$$

and also noting the initial conditions

$$u_i^0 = \frac{1}{h} \int_{x_{i-\frac{1}{2}}}^{x_{i+\frac{1}{2}}} u_0(x) \, dx,$$

we have

$$\begin{aligned} \int_0^\infty \int_{\mathbb{R}} u_h(x, t+\tau) \left(\frac{v_h(x, t+\tau) - v_h(x,t)}{\tau} \right) dxdt \\ + \int_0^\infty \int_{\mathbb{R}} f_h(x,t) \left(\frac{v_h(x+\frac{h}{2}, t) - v_h(x-\frac{h}{2}, t)}{h} \right) dxdt \\ + \int_{\mathbb{R}} u_0 v_h(x,0) \, dx = 0. \end{aligned} \tag{9.35}$$

If u_h converges to u as in (9.33), from the preceding equation (9.35) one can proof, using standard theorems from integration theory, that the limit function u satisfies (9.34) in the limit $h \to 0$. (Note that the terms in the brackets are consistent difference approximations to v_t and v_x.) ■

9.6 Boundary Conditions

We have not so far discussed boundary conditions. For hyperbolic problems it is important to set boundary conditions consistent with the propagation of information established by the

characteristics. Recall first the simplest case, the scalar advection equation, on a finite domain, for example,

$$u_t + au_x = 0, \qquad x \in (0,1),\ t > 0, \qquad a \in \mathbb{R}.$$

From the analysis in section 5.3 we recall that for $a > 0$ we have to set boundary conditions at the "left" boundary $x = 0$, but we must not specify the solution at $x = 1$. (If $a < 0$, the situation is opposite.)[6]

For nonlinear scalar problems, i.e. (8.15) the situation is similar. Recall that the characteristics are still straight lines, and thus one must still prescribe boundary conditions whenever characteristics enter the domain. There may be situations, where boundary conditions must be prescribed at both boundaries, or no boundary conditions may be prescribed at all. There may also be cases, where boundary data must be prescribed on some portions on the boundary, but not on others.

Example 9.3 Consider the Burgers' equation

$$\begin{aligned} u_t + \left(\frac{u^2}{2}\right)_x &= 0, \qquad x \in (0,1),\ t > 0. \\ u(x,0) &= u_0(x), \qquad x \in [0,1] \end{aligned}$$

Convince yourself that for the initial conditions

$$u_0(x) = x - \frac{1}{2}$$

all characteristics are outgoing, and we must not prescribe boundary conditions either at $x = 0$, or at $x = 1$. Conversely, for the initial conditions

$$u_0(x) = \frac{1}{2} - x$$

the situation is opposite. All characteristics run towards the interior of the domain.

For nonlinear systems the picture is more complicated, and we will not discuss the issue here.

Implementation of the boundary conditions is straightforward, but different from the finite-difference schemes. The latter have been extensively discussed in previous chapters. For finite-volume schemes we implement boundary conditions as follows. Consider again a scalar one-dimensional problem. Let us assume we need to set boundary conditions at the left boundary, and boundary data is given as $u = \psi(t)$ there. Let us consider a finite-volume scheme, and assume that the first cell has index $i = 1$. Then we have for that cell

$$u_1^{n+1} = u_1^n - \frac{\tau}{h}\left(\hat{f}(u_1, u_2) - f(\psi(t))\right),$$

meaning we evaluate the flux function with the boundary conditions instead of the numerical solution.

9.7 Nonlinear Systems of Conservation Laws

Recall first the linear system

$$\mathbf{u}_t + A\mathbf{u}_x = 0,$$

[6] For linear systems (cf. section 7), the situation is similar. In fact, after a system, such as eq. (7.1) has been transformed to characteristic variables (e.g. eq. 7.3), the problem decomposes into independent advection equations, see (7.5). Consequently, we can set boundary conditions for each of the characteristic variables, just like for the scalar linear advection equation.

where $A \in \mathbb{R}^{m\times m}$ is a constant matrix, and recall the upwind scheme for linear systems (cf. (7.7) in Chapter 7), which is obtained by applying the upwind scheme to each advection equation that results from spectral decomposition of the linear system. (Recall that such a decomposition is always possible by definition of a linear hyperbolic system.) We write this scheme as

$$\mathbf{u}_i^{n+1} = \mathbf{u}_i^n - \frac{\tau A}{2h}\left(\mathbf{u}_{i+1}^n - \mathbf{u}_{i-1}^n\right) + \frac{\tau|A|}{2h}\left(\mathbf{u}_{i-1}^n - 2\mathbf{u}_i^n + \mathbf{u}_{i+1}^n\right).$$

(See Chapter 7 for pertaining information.) Setting $\mathbf{f}(\mathbf{u}) = A\mathbf{u}$, and defining a numerical flux

$$\hat{\mathbf{f}}_{i+\frac{1}{2}} = \frac{1}{2}\left(\mathbf{f}(\mathbf{u}_{i+1}^n) + \mathbf{f}(\mathbf{u}_i^n)\right) - \frac{|A|}{2}\left(\mathbf{u}_{i+1}^n - \mathbf{u}_i^n\right), \tag{9.36}$$

we can write the scheme as

$$\mathbf{u}_i^{n+1} = \mathbf{u}_i^n - \frac{\tau}{h}\left(\hat{\mathbf{f}}_{i+\frac{1}{2}} - \hat{\mathbf{f}}_{i-\frac{1}{2}}\right).$$

Thus, we note that the difference scheme has a straightforward interpretation as a finite-volume scheme. The upwind method is hardly the only way of defining a reasonable scheme for systems. Instead of using $|A|$ to scale the diffusion terms we may consider a scalar diffusion coefficient α, such that

$$\hat{\mathbf{f}}_{i+\frac{1}{2}} = \frac{1}{2}\left(\mathbf{f}(\mathbf{u}_{i+1}^n) + \mathbf{f}(\mathbf{u}_i^n)\right) - \frac{\alpha}{2}\left(\mathbf{u}_{i+1}^n - \mathbf{u}_i^n\right)$$

where we can define α in such a way that the scheme has at least as much diffusion as the upwind scheme by setting $\alpha = \max_{1\le p\le m}|\lambda_p|$, where λ_p is the pth eigenvalue of the matrix A.

For general nonlinear system

$$\mathbf{u}_t + \mathbf{f}(\mathbf{u})_x = 0,$$

we consider as a general flux function

$$\hat{\mathbf{f}}_{i+\frac{1}{2}} = \frac{1}{2}\left(\mathbf{f}(\mathbf{u}_{i+1}^n) + \mathbf{f}(\mathbf{u}_i^n)\right) - \frac{B^n}{2}\left(\mathbf{u}_{i+1}^n - \mathbf{u}_i^n\right)$$

where a choice similar to (9.36) leads to

$$B^n = |A(\mathbf{u}_{i+1}^n, \mathbf{u}_i^n)|,$$

and $A(\mathbf{u},\mathbf{v})$ is a suitable linearization of the flux function, i.e. a consistent approximation to the Jacobian, such that $A(\mathbf{u},\mathbf{u}) = \mathbf{f}_{\mathbf{u}}$. Concrete Examples will be discussed in the tutorial. Alternatively, a scalar diffusion coefficient may be written

$$B^n = \alpha(\mathbf{u}_{i+1}^n, \mathbf{u}_i^n)I$$

where I is the identity matrix, and $\alpha(\mathbf{u},\mathbf{v})$ approximates the spectral radius (i.e. the eigenvalue with maximum magnitude) of the flux Jacobian.

In multiple dimensions, a similar method may be used. Consider the conservation law

$$\mathbf{u}_t + \sum_{i=1}^{d}(\mathbf{f}(\mathbf{u}))_{x_i} = 0,$$

and the corresponding finite volume scheme of Section 9.2,

$$\mathbf{u}_i^{n+1} = \mathbf{u}_i^n - \frac{\tau}{|T_i|}\sum_{k\in\mathcal{N}_i}\hat{\mathbf{f}}(\mathbf{u}_i^n, \mathbf{u}_k^n; n_{ik})|S_{ik}|.$$

We may consider the general flux function

$$\hat{\mathbf{f}}(\mathbf{u},\mathbf{v};n) = \frac{1}{2}\left(\mathbf{f}(\mathbf{u})\cdot n + \mathbf{f}(\mathbf{v})\cdot n\right) - \frac{B(\mathbf{u},\mathbf{v};n)}{2}\left(\mathbf{v}-\mathbf{u}\right),$$

where now

$$B = |A(\mathbf{u},\mathbf{v};n)|$$

is a consistent linearization of the normal flux function, i.e. $A(\mathbf{u},\mathbf{u};n) = \mathbf{f}_u\cdot n$. Alternatively we may use $B = \alpha(\mathbf{u},\mathbf{v};n)I$, where α is an approximation of the spectral radius of $\mathbf{f}_{\mathbf{u}}\cdot n$.

9.8 Further Reading

The Lax-Wendroff Theorem is discussed and proved in Godlewski and Raviart [7]. Barth and Ohlberger [2] discuss results summarized here, as well as further extensions, such as proofs of convergence and higher order extensions. (Theorems are generally not proved in [2]. But it is nevertheless a good reference, because the most relevant results are summarized, and references are given where proofs can be found.)